1,4-cyclohexadiene reacting with 96 percent sulfuric acid

Rebecca Thomas

Southern Adventist University

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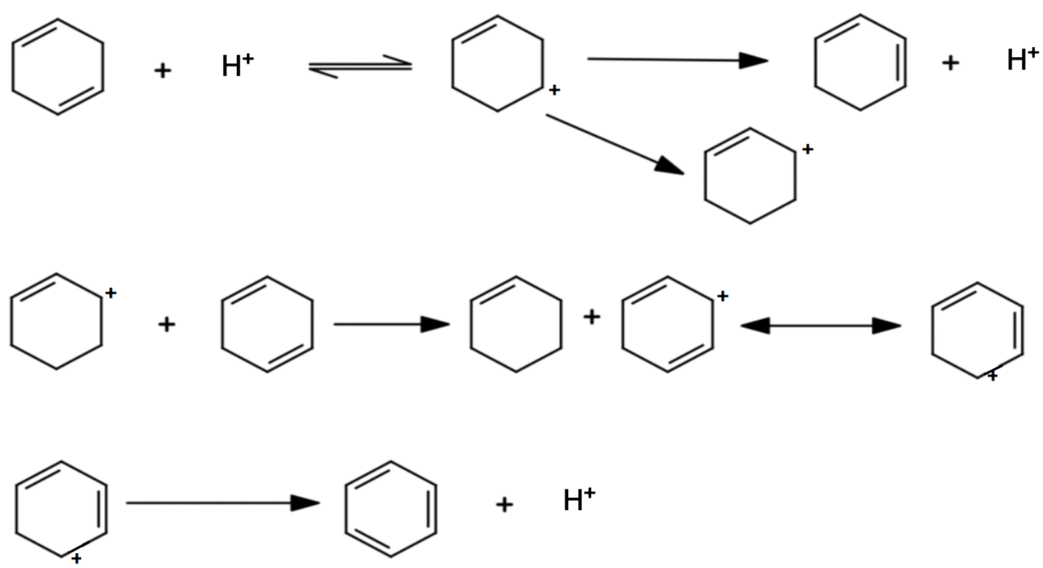
Abstract

This research aims to determine the order, the rate constant, and the activation energy of the reaction of 1,4-cyclohexadiene with 96% sulfuric acid. The proposed mechanism occurs when a cycloalkenyl carbocation undergoes a hydride shift which results in the formation of a cyclodienylic cation (refer to the proposed mechanism). The cycloalkenyl carbocation absorbs strongly at around 320 nanometers and the cyclodienylic cation absorbs strongly at around 385 nanometers[1,2]. This allows the concentrations of both carbocations to be tracked using ultraviolet visible spectroscopy. The order, the rate constant, and the activation energy for the reaction are found by tracking the change in absorbance at both 320 and 385 nanometers at various temperatures. The activation energy found for 320 nanometers is determined to be 72.2 kJ/mol. The activation energy found for 385 nanometers is determined to be 20.7 kJ/mol.

Background

Cycloalkenyl cations are formed using concentrated sulfuric acid and a cycloalkene and are known to exhibit ultraviolet absorbance with a single symmetrical shape where the λ_{max} is 300 nanometers and the molar absorptivity is around 10^4 [1,3]. While the lifetime of most carbocations is in the range of 91 nanoseconds, cycloalkenyl cations have been observed to remain unchanged in highly concentrated sulfuric acid after one hour as evidenced by N.M.R. spectra[3]. Due to its high stability and lambda max within the ultraviolet range, equipment needed to study cycloalkenyl cations can be standard laboratory equipment. The proposed hypothesis is based on the proposed mechanism is shown below. In step one of the mechanism, 1,4-cyclohexadiene is protonated and can form a secondary carbocation. In the second step, it is proposed that a hydride shift will occur and an allylic cycloalkenyl cation will form. In the final step of the mechanism, it is proposed that a benzene ring will form.



(1)

(2)

(3)

It is expected to see evidence of a cycloalkenyl cation with a single symmetrical shape where the λ_{max} is 300 nanometers and the molar absorptivity is around 10^4. The rate limiting step is expected to be the first step because the stable cyclodiene is changing to a less stable secondary carbocation[4]. It is expected that the activation energy for the proposed reaction to be around 69.2 kJ/mol [5]. The activation energy would decrease with an increase in sulfuric acid concentration. The purpose of this experiment is to find the activation energies and to do this, several ultraviolet spectra will be conducted at a constant temperature and analyzed. Several experiments will also be run at different temperatures. This will help determine the rate constant from which the activation energies can be found as suggested by the Arrhenius equation (equation (1)). Several rate constants will be found at different temperatures.

$$k = Ae^{-E_a/RT} \; or \ln k = \left(\frac{-E_a}{RT}\right) + \ln A \tag{1}$$

where k = the rate constant, A = the frequency or pre-exponential factor, E_a= the activation energy, R = the ideal-gas constant, and T = temperature

Experimental Method

The method used was time-series absorbance ultraviolet visible spectrophotometry (company: Shimadzu, model: UV-2600, software: UV probe v.252) with a scanning range of 200 to 450 nanometers. Multiple scans were conducted over time at different temperatures ranging from 30 to 38 degrees Celsius to look for the activation energy and characteristics of the intermediates in the mechanism.

When creating the 1,4-cyclohexadine dilution, approximately 0.1 g of 1,4-cyclohexadiene (company: Aldrich, purity: 97%, catalog number: 125415-5ML) was dissolved in approximately 100 mL of methanol (company: Acros, purity: 99.8%, catalog number: 176840025). This solution was then injected with a 50-microliter syringe (company: Hamilton, model number: 705 N) into standard closed cap cuvettes (company: Starna Cells, Inc., model: Quartz, 10mm path, catalog number: 1-Q-10-GL14-C) containing 96% sulfuric acid (company: Fisher, catalog number: A300-212). For the total experiment, approximately 50 mL of sulfuric acid was used.

The reactions were carried out at various temperatures that were held steady by a temperature control unit (company: Fisher Scientific Isotemp 3016). Glassware and sundry items common to the laboratory were also used. Data was obtained and was kinetically analyzed so that characteristics of the intermediates in the mechanism could be observed and that the order, rate, and activation energy were obtained for the reactions at various temperatures.

Results

In this experiment, spectra were collected at varying temperatures within the range of 30 to 38 degrees Celsius. An example of one such spectra is shown in Figure 1 below. For Figure 1, values of absorbance over wavelength plotted and at 220 nanometers, 320 nanometers, and 385 nanometers peaks were observed. Other spectra collected were similar, however, each spectrum did not always show the same peaks consistently with increased temperature.

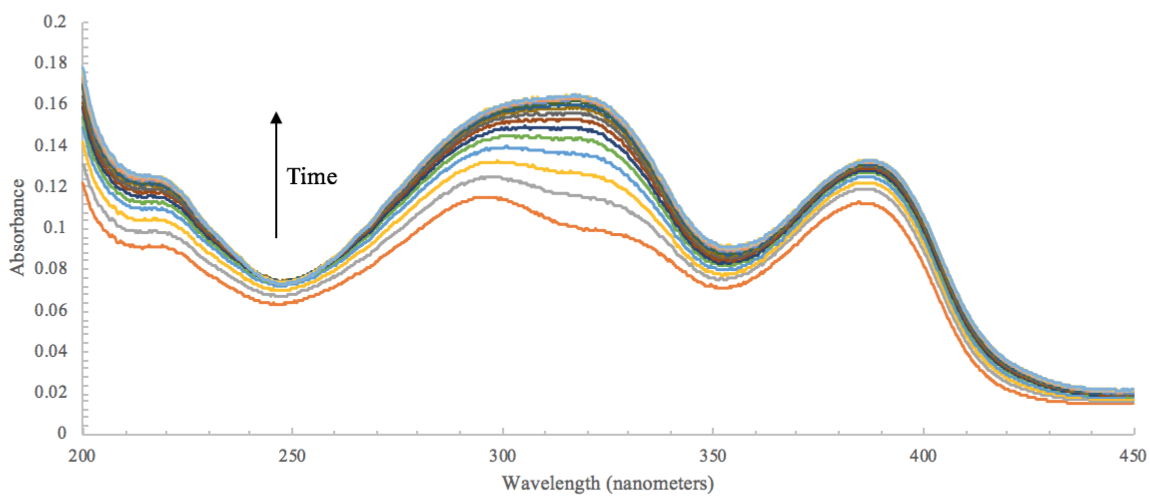


Figure 1. Time-series absorbance spectra taken at 35.0 degrees Celsius when 21.59 microliters of 0.01621 M 1,4 cyclohexadiene in methanol was added to 3.5 milliliters of 96% sulfuric acid. Spectra were recorded every 120 seconds. Peaks were identified at around 220 nanometers, 320 nanometers, and 385 nanometers.

A peak at 220 nanometers was observed for 32 degrees Celsius, 35 degrees Celsius, and 38 degrees Celsius. As seen in Figure 2, values of absorbance were plotted over time for spectra conducted at those three temperatures.

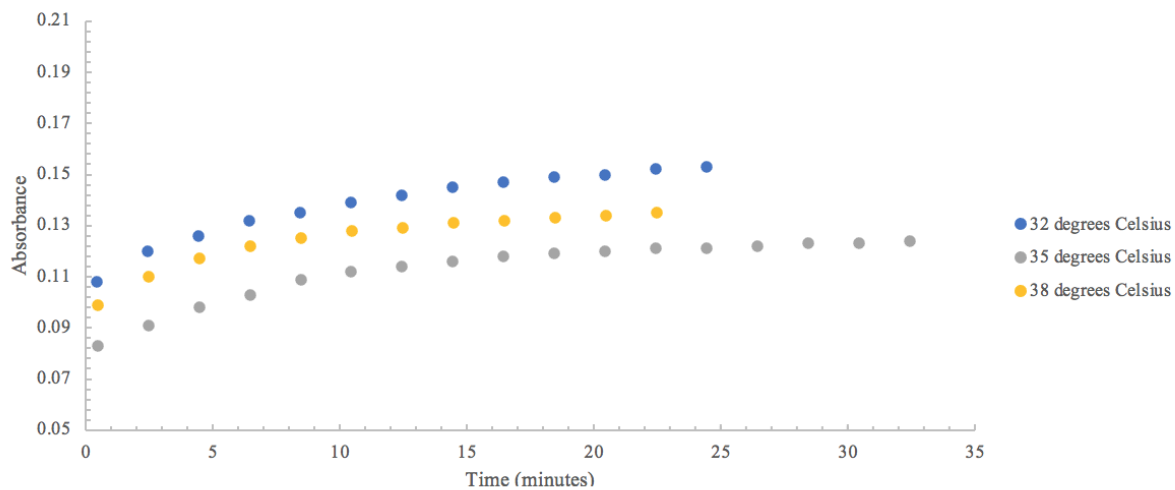


Figure 2. Absorbance over time plot at 220 nanometers for 32°C, 35°C, and 38°C.

A peak at 255 nanometers was observed for 32 degrees Celsius and 38 degrees Celsius. As seen in Figure 3, values of absorbance were plotted over time for spectra conducted at those two temperatures.

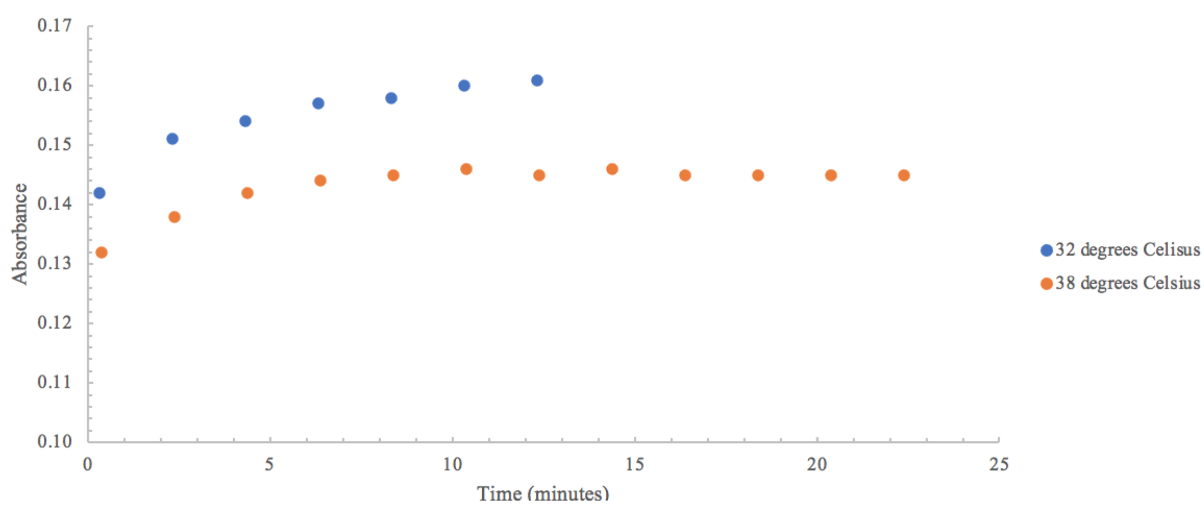


Figure 3. Absorbance over time plot at 255 nanometers for 32°C and 38°C.

A peak at 320 nanometers was observed for 32 degrees Celsius, 35 degrees Celsius, and 38 degrees Celsius. As seen in Figure 4, values of absorbance were plotted over time for spectra conducted at those three temperatures.

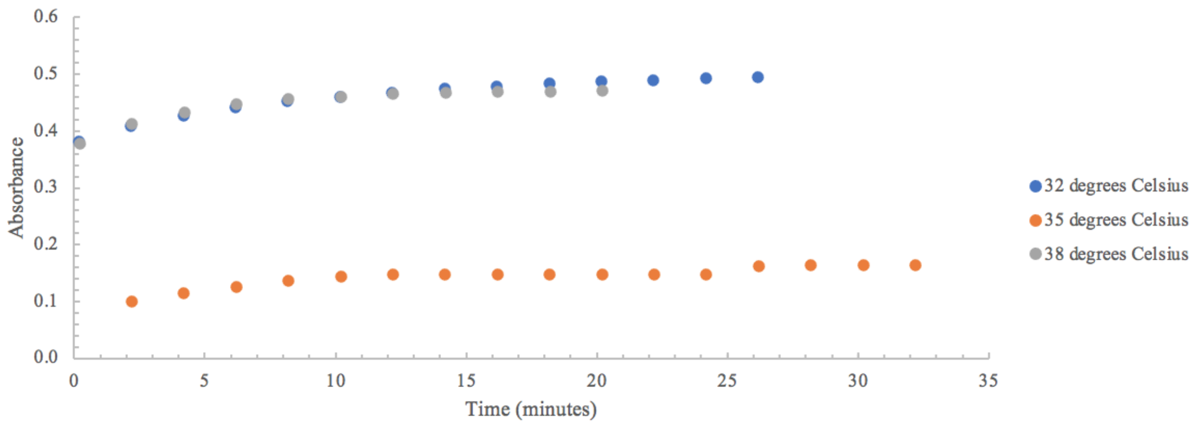


Figure 4. Absorbance over time plot at 320 nanometers for 32°C, 35°C, and 38°C.

A peak at 385 nanometers was observed for 30 degrees Celsius, 32 degrees Celsius, 34 degrees Celsius, 35 degrees Celsius, and 38 degrees Celsius. As seen in Figure 5, values of absorbance were plotted over time for spectra conducted at those three temperatures.

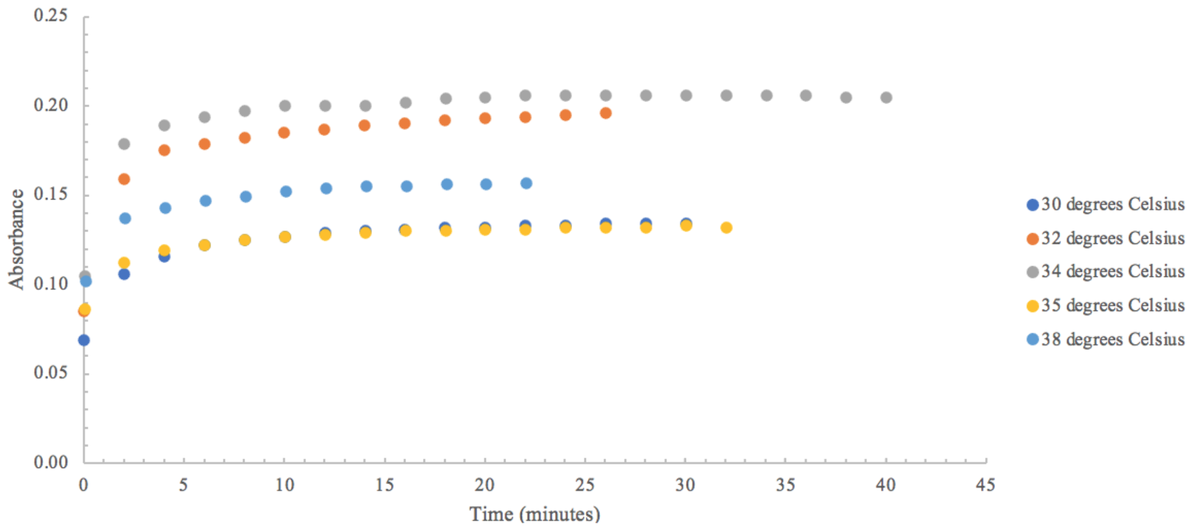


Figure 5. Absorbance over time plot at 385 nanometers for 30°C, 32°C, 34°C, 35°C, and 38°C.

For 220 nanometers, values of the natural log of the absorbance were plotted over time for 32 degrees Celsius, 35, degrees Celsius, and 38 degrees Celsius as seen in Figure 6. The trendlines and their equations and correlation values are also shown.

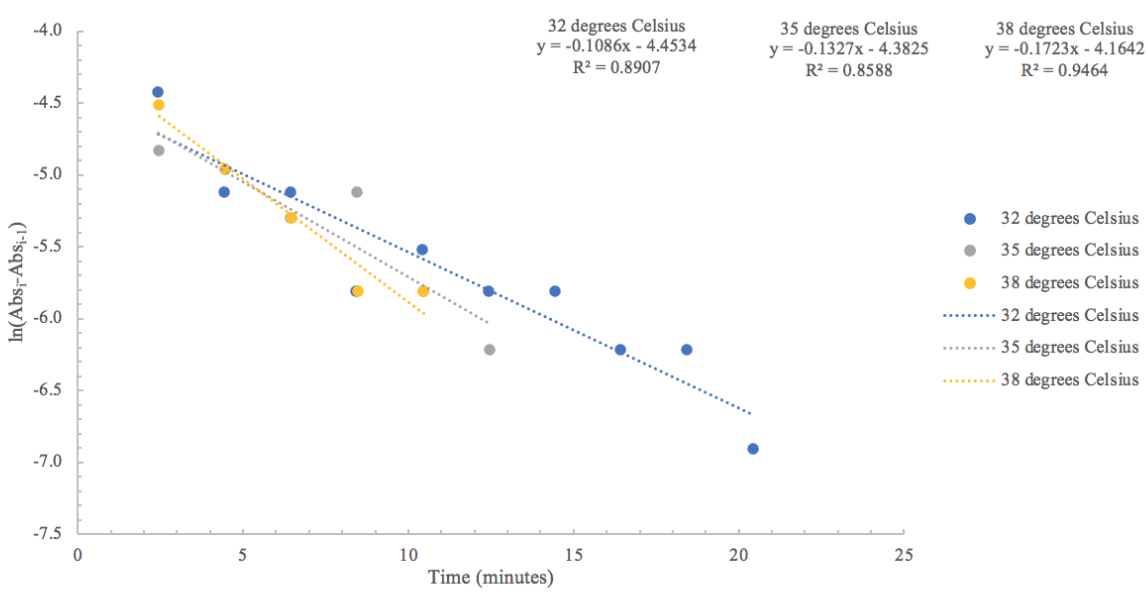


Figure 6. The natural log of the absorbance over time plot at 220 nanometers for 32°C, 35°C, and 38°C.

For 255 nanometers, values of the natural log of the absorbance were plotted over time for 32 degrees Celsius and 38 degrees Celsius as seen in Figure 7. The trendlines and their equations and correlation values are also shown.

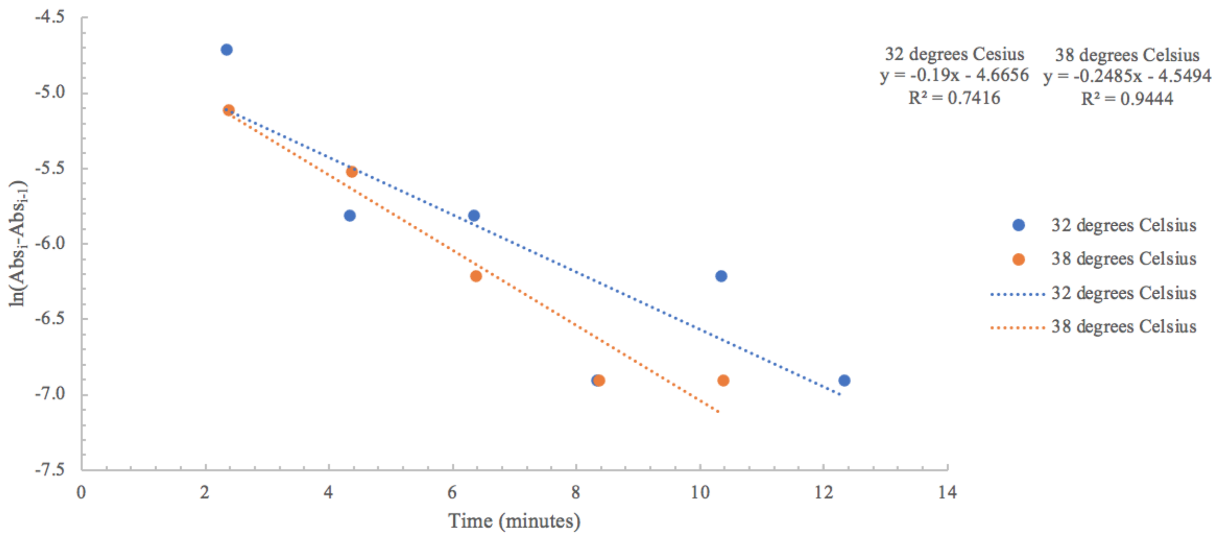


Figure 7. The natural log of the absorbance over time plot at 255 nanometers for 32°C, and 38°C.

For 320 nanometers, values of the natural log of the absorbance were plotted over time for 32 degrees Celsius, 35, degrees Celsius, and 38 degrees Celsius as seen in Figure 8. The trendlines and their equations and correlation values are also shown.

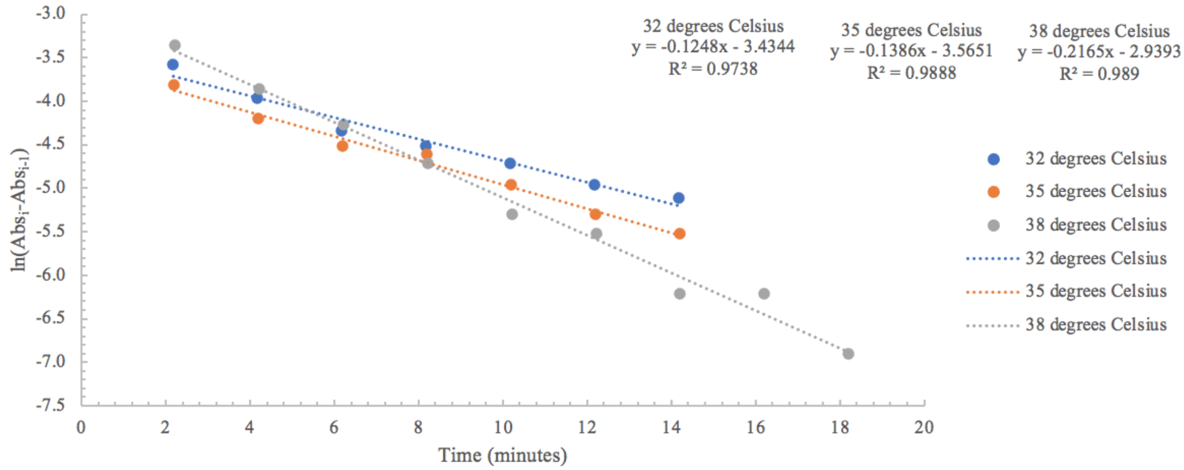


Figure 8. The natural log of the absorbance over time plot at 320 nanometers for 32°C, 35°C, and 38°C.

For 385 nanometers, values of the natural log of the absorbance were plotted over time for 30 degrees Celsius, 32, degrees Celsius, 34 degrees Celsius, 35, degrees Celsius, and 38 degrees Celsius as seen in Figure 9. The trendlines and their equations and correlation values are also shown.

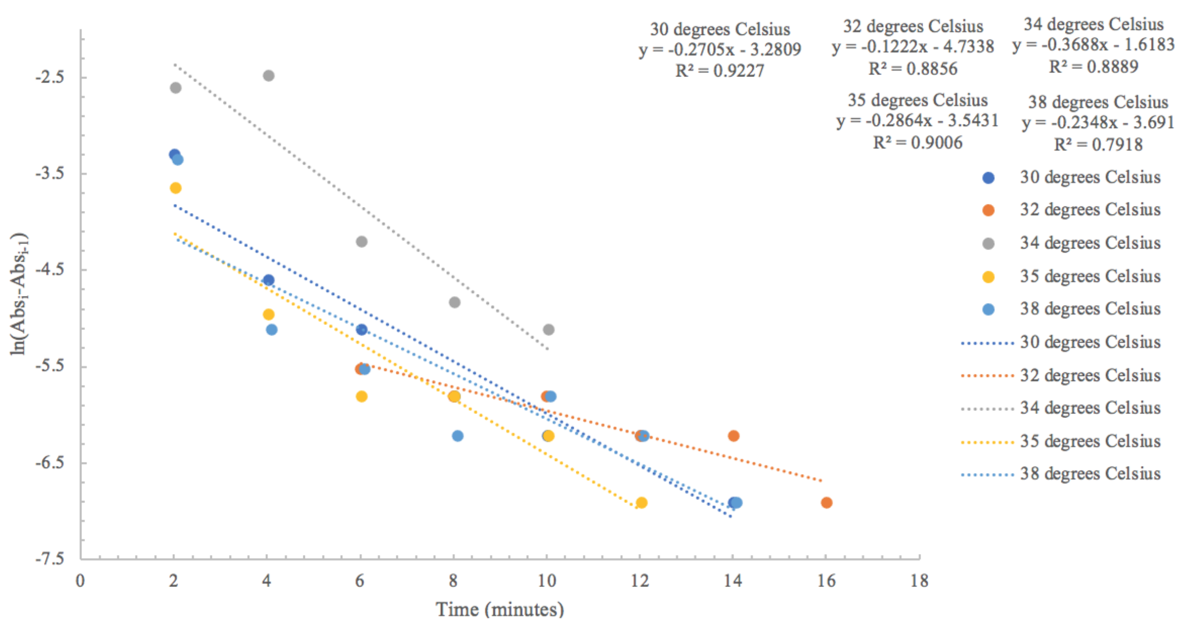


Figure 9. The natural log of the absorbance over time plot at 385 nanometers for 30°C, 32°C, 34°C, 35°C, and 38°C.

The observed k values, taken from the slopes of the trendlines from Figures 7- 10, for each temperature and their respective peaks were tabulated in Table 1 below.

Table 1. The k values associated with each temperature and wavelength.

Wavelengths (nanometers)	Temperatures (Celsius) 30	32	34	35	38
220	-	0.1086	-	0.1327	0.1723
255	-	0.19	-	-	0.2485
320	-	0.1248	-	0.1386	0.2165
385	0.2705	0.1222	0.3688	0.2864	0.2348

Finally, the observed rate constants over the inverse temperature values were plotted for 220 nanometers, 320 nanometers, and 385 nanometers as seen in Figure 10. The trendlines and their equations and correlation values are also shown.

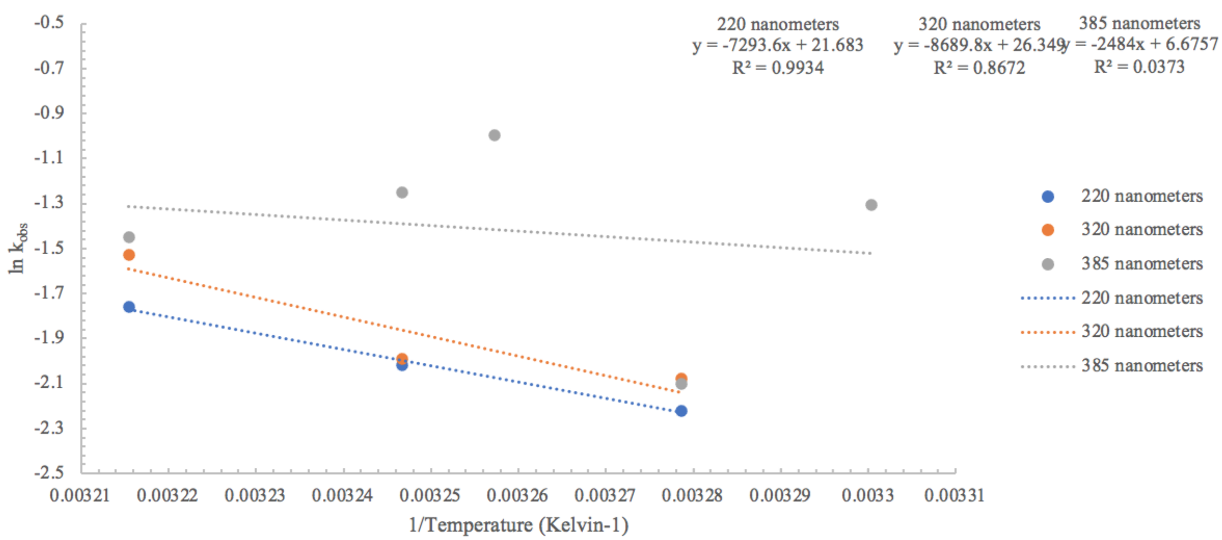


Figure 10. An Arrhenius plot of the natural log of k_{obs} over the inverse temperature at 220 nanometers, 320 nanometers, and 385 nanometers.

From the plot in Figure 10, the activation energies for those wavelengths were calculated as seen in Table 2 below.

Table 2. The activation energies and their corresponding wavelengths.

Wavelength (nanometers)	Activation Energy (kilojoules/mole)
220	60.7
320	72.2
385	20.7

Discussion

The peaks observed from the spectra collected at 250 nanometers, 320 nanometers, and 380 nanometers correspond to intermediates within the proposed mechanism. The peak at 250 nanometers is believed to correspond to the presence of 1,3-cyclohexadiene, unreacted 1,4-cyclohexadiene, or a benzene ring[6]. The peak at 320 nanometers is believed to correspond to the presence of the cycloalkenyl cation shown in the proposed mechanism[1]. The peak at 385 nanometers is believed to correspond to the presence of the cyclodienylic carbocation shown in the proposed mechanism[3]. The activation energy found for 320 nanometers was 72.2 kJ/mol which corresponds to previous findings of the activation energy for cyclohexanol which was around 69.2 kJ/mol [5].

During the experiment, not all four peaks (at 220 nanometers, 255 nanometers, 320 nanometers, and 385 nanometers) showed up for each temperature. It is suspected that these findings are related to the reproducibility of the injection of the 1,4-cyclohexadiene dilution into the sulfuric acid3. This would be good to explore further in future work.

References

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